

C L VIBIN CARIAPPA

AI/ML Engineer | Generative AI Engineer | Backend Developer

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PROFESSIONAL SUMMARY

Artificial Intelligence Engineer specializing in building Agentic AI systems, Generative AI application, Large Language Models, Retrieval-Augmented Generation, and intelligent automation platforms. Hands-on experience designing and deploying autonomous AI agents, context-aware LLM workflows, semantic search pipelines, and enterprise AI applications using Python, LangChain, LangGraph, OpenAI APIs, Hugging Face Transformers, PyTorch, and vector databases. Experienced in integrating AI models with external tools, REST APIs, databases, and production systems while applying prompt engineering, memory management, retrieval optimization, model evaluation, and scalable AI architecture principles.

TECHNICAL SKILLS

Programming Languages	Retrieval Systems & Vector Databases
- Python - Java - JavaScript - SQL	- Qdrant, FAISS, Pinecone - Semantic Search, Hybrid Search - Reranking Pipelines - Context Retrieval, Context-Aware Response Generation
AI/ML & Generative AI	Backend & AI Infrastructure
- Supervised & Unsupervised Learning, Neural Networks, - Natural Language Processing, Computer Vision - CNNs, RNNs, Transformers - LLMs, RAG, Prompt Engineering, Agentic AI Systems - Multimodal AI, Multi-Agent Architectures - Feature Engineering, Context Engineering - Model Optimization, Model Evaluation - LLM Orchestration, Tool Calling & Function Calling	- FastAPI, REST APIs - Express.js, Next.js - Backend API Development, Backend Integrations - End-to-End AI Pipeline Architecture - AI Application Architecture - AI Model Deployment, Scalable AI Systems
Frameworks & Libraries	Databases & Data Engineering
- NumPy, Pandas, Matplotlib - PyTorch TensorFlow, Scikit-learn - LangChain, LangGraph, LangSmith - CrewAI, Hugging Face Transformers	- PostgreSQL, Prisma - Vector Databases - Data Processing Pipelines - Data Preprocessing, Data Analysis
Model Training & Inference	Cloud, DevOps & MLOps
- Fine-Tuning, Transformer Fine-Tuning - Supervised Fine-Tuning, Parameter-Efficient Fine-Tuning - LoRA / QLoRA - GPU Optimization, Model Inference Optimization	- AWS, GCP - Docker, Kubernetes - Git, GitHub Actions, CI/CD Pipelines - Production AI Systems, Cost Optimization Systems

PROFESSIONAL EXPERIENCE

Mitcon Consultancy & Engineering Service | AI Fullstack Intern

Apr 26 - Present

Enterprise Data Intelligence & Impact Management Platform

- Developed an enterprise-grade data intelligence platform for beneficiary management, impact monitoring, document processing, and analytics-driven decision support.
- Designed and developed backend services and REST APIs supporting data management workflows, automated file ingestion, analytics dashboards, reporting systems, and performance tracking modules.
- Built automated document processing and data aggregation pipelines for extracting, organizing, and analyzing structured and unstructured data, reducing manual reporting and operational efforts.

Tech Stack: Python, Node.js, Express.js, PostgreSQL, REST APIs, Data Processing Pipelines

Sash.AI | AI Engineer Trainee

Feb 2026 - Mar 2026

An AI Podcast Interviewer

- Contributed to the development of an LLM-powered AI Podcast Interviewer capable of generating dynamic, context-aware interview conversations.
- Designed prompt engineering workflows for persona-driven questioning, follow-up generation, and topic-aware dialogue management.
- Evaluated generated responses for relevance, coherence, and conversational quality using custom testing frameworks.
- Collaborated on improving interview flow, question diversity, and contextual understanding through iterative prompt optimization.

Tech Stack: Python, LLMs, NLP, FastAPI, RAG, Vector Embeddings, LangChain

PROJECTS

RAG Based AI Agent for Document Q&A

- Built an advanced RAG with orchestration for multi-hop queries and explainable answers.
- Designed agent orchestrator using type-safe prompts and chain-of-thought prompting
- Improved accuracy to 92%; users trusted answers due to source citations; reduced hallucinations by 80%

Tech Stack: Python, Qdrant, OpenAI/LLaMA, Vector Embeddings, FAISS

GPT-Style Language Model Implementation (nanoGPT)

- Implemented a GPT-style transformer architecture from inspired by Andrej Karpathy’s nanoGPT implementation.
- Developed tokenization, self-attention, positional embeddings, and training workflows.
- Trained language models using next-token prediction objectives using the “Project Gutenberg Corpus” dataset.
- Explored optimization techniques and transformer internals.

Tech Stack: PyTorch, Deep Learning, Transformers

Localized LLM Fine-Tuning - BoDmagh

- Fine-tuned transformer-based language models on domain-specific conversational datasets.
- Developed preprocessing, tokenization, and supervised training pipelines.
- Performed hyperparameter tuning and model evaluation experiments.
- Improved domain adaptation and contextual understanding through targeted fine-tuning.

Tech Stack: Python, PyTorch, Hugging Face Transformers, PEFT

AI-Powered Resume Analyzer & ATS Scorer

- Developed an AI-driven resume analysis platform leveraging NLP and transformer embeddings.
- Implemented semantic matching between resumes and job descriptions.
- Built ATS scoring, resume parsing, and skill extraction pipelines.
- Generated AI-assisted recommendations for resume optimization.

Tech Stack: Python, NLP, Transformers, FastAPI

Sign.mt – Real-Time Communication Platform for specially abled

- Ranked Runner-Ups among the Inter - Department project exhibition
- Developed real-time backend infrastructure using WebSockets (Socket.IO) for bidirectional low-latency communication
- Implemented event-driven architecture handling 1000+ concurrent connections with optimized message broadcasting
- Designed fault-tolerant connection management and automatic reconnection mechanisms ensuring 99.9% uptime

Tech Stack: Node.js, Express, Socket.IO, WebSockets, Event-Driven Architecture

ACHIEVEMENTS

Runner-Up - Inter Department Project Exhibition CIT, Kodagu	Jan 2026
Ranked Top 4 / 140+ teams - Srinathon Hackathon SIT, Manglore	Nov 2024

CERTIFICATES

- Generative AI for All - Infosys Springboard
- Machine Learning Implementation - Infosys Springboard
- Software Engineering - Electronic Arts
- GenAI Data Analytics - Tata Group
- Cybersecurity Analytics - Tata Group

EDUCATION

Coorg Institute of Technology	
B.E. CSE (AI & ML)	2022 - 2026
CGPA: 8.41	
St. Joseph’s P U College	
PCMB: 83.5%	2020 - 2022
Kendriya Vidyalaya Kodagu	
SSLC: 86.5 %	2019 - 2020